

Obstacle-Aware Leader-Follower Formation Control via Deep Q-Networks

Omitted due to Blind Review¹, To be included if accepted²

Abstract—This paper presents a hybrid control architecture for leader-follower formation control of non-holonomic mobile robots operating in obstacle-rich environments. The proposed approach combines Deep Reinforcement Learning for the leader robot with classical proportional control for the follower, effectively decoupling navigation complexity from formation maintenance. The leader employs a Deep Q-Network (DQN) trained to navigate toward goal positions while avoiding static obstacles. The follower tracks a virtual target point positioned behind the leader using a computationally efficient proportional controller, maintaining formation integrity. The architecture is validated in a challenging scenario requiring both robots to traverse a narrow 1.0m passage while maintaining 0.3m formation distance. Experiments across three initial configurations of increasing difficulty demonstrate successful goal convergence and reveal emergent coordinated behavior: the learned policy autonomously generates maneuvers that create clearance for the follower in constrained spaces, despite being trained as a single-agent system. Results confirm that the hybrid architecture effectively balances the adaptive capabilities of learning-based methods with the reliability of classical control, making it suitable for heterogeneous multi-robot systems.

I. INTRODUCTION

Multi-robot systems (MRS) have emerged as a transformative paradigm in robotics, offering enhanced redundancy, parallel task execution, and scalability compared to single-agent solutions [1]. Among the various coordination strategies, leader-follower architectures have gained widespread adoption due to their intuitive structure and ease of implementation [2]. In this paradigm, one or more designated leaders determine the group's trajectory, while followers maintain prescribed spatial relationships relative to the leaders. The applications of leader-follower formations span numerous domains, such as industrial logistics [3], space exploration [4], precision agriculture [5], and in defense and search-and-rescue operations [6].

Despite recent advances, coordinating non-holonomic robots in cluttered environments remains challenging. The kinematic constraints inherent to differential-drive platforms restrict instantaneous lateral motion, complicating precise position control [7]. Furthermore, obstacle avoidance requires the leader to deviate from optimal paths, forcing followers to react dynamically while maintaining formation integrity.

Classical approaches to this problem include Artificial Potential Fields (APF) [8], which provide elegant reac-

tive behaviors but suffer from local minima in complex obstacle configurations. Model Predictive Control (MPC) offers optimal trajectory planning with constraint satisfaction [9], yet computational demands may exceed real-time requirements for resource-constrained platforms. Recently, Deep Reinforcement Learning (DRL) has emerged as a promising paradigm for robot navigation, enabling agents to learn policies directly from sensory experience [10]. The landmark work by Mnih et al. [11] demonstrated that Deep Q-Networks could achieve human-level performance across diverse Atari games, inspiring applications in continuous control domains including mobile robotics [12].

This paper addresses leader-follower formation control in environments with static obstacles through a hybrid architecture. The leader robot employs a DQN-based policy for autonomous navigation and obstacle avoidance, while the follower utilizes a classical proportional controller for formation tracking. This design philosophy assigns computational complexity to the leader—typically equipped with superior sensing and processing capabilities—while enabling lightweight followers to maintain formation through simple, robust control laws. The main contributions of this work is the proposal of a hybrid control architecture combining DRL-based navigation for the leader with proportional tracking control for the follower, suitable for heterogeneous robot teams with obstacle avoidance capability.

The remainder of this paper is organized as follows. Section II presents the theoretical foundations and reviews related work. Section III formulates the problem mathematically. Section IV details the proposed hybrid control architecture. Section V-A presents simulation results, and Section VI concludes the paper.

II. FUNDAMENTALS AND RELATED WORK

A. Theoretical Foundation

The agents considered in this work are differential-drive mobile robots subject to non-holonomic constraints. The kinematic model follows the unicycle representation. Let $q = [x, y, \theta]^T$ denote the robot's pose in the global inertial frame, where (x, y) represents position and θ orientation. The kinematic equations are given by:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos(\theta) & 0 \\ \sin(\theta) & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix} \quad (1)$$

where v and ω denote linear and angular velocities, respectively. The non-holonomic constraint restricts lateral motion,

*to be included

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satisfying:

$$\dot{x} \sin(\theta) - \dot{y} \cos(\theta) = 0 \quad (2)$$

This constraint adds complexity to follower control, particularly during sharp leader turns [7].

The unicycle model (1) is underactuated: three degrees of freedom are controlled by two inputs. A common technique to address this limitation involves controlling a virtual point displaced from the wheel axis center [13]. Consider a point (x_p, y_p) located at distance L ahead of the robot center:

$$x_p = x + L \cos(\theta), \quad y_p = y + L \sin(\theta) \quad (3)$$

The dynamics of this virtual point yield a fully linearized system when $L \neq 0$, enabling standard linear control techniques for position regulation. This concept extends naturally to leader-follower formations, where the follower tracks a virtual target behind the leader, treating the formation as a flexible mechanical linkage [14].

Reinforcement Learning (RL) formulates sequential decision-making as a Markov Decision Process (MDP), defined by the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ the action space, P the transition probability, R the reward function, and $\gamma \in [0, 1)$ the discount factor [15]. The objective is to find a policy π maximizing expected cumulative reward.

For high-dimensional state spaces, tabular Q-learning becomes intractable. Deep Q-Networks (DQN), introduced by Mnih et al. [11], approximate the Q-function using deep neural networks parameterized by weights θ . Two mechanisms ensure training stability:

- **Experience Replay:** Transitions (s, a, r, s') are stored in a replay buffer and sampled uniformly during training, breaking temporal correlations between consecutive samples [16].
- **Target Network:** A separate network with parameters θ^- , updated periodically, computes target Q-values, preventing oscillations from rapidly changing targets.

The loss function minimized during training is:

$$\mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} \left[\left(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta) \right)^2 \right] \quad (4)$$

B. Related Work

Formation control has been extensively studied, with three predominant paradigms: leader-follower, virtual structure, and behavior-based approaches [2]. The leader-follower strategy, adopted in this work, offers simplicity and natural scalability but requires reliable communication and places significant responsibility on the leader [14].

Oh et al. [2] provide a comprehensive survey categorizing formation control into position-based, displacement-based, and distance-based methods. Traditional approaches include separation-bearing control [17], which assumes obstacle-free environments, and potential field methods [18], which can trap robots in local minima. Recent advances incorporate

Model Predictive Control for constraint satisfaction [9] and consensus-based algorithms for distributed coordination [19].

The application of Deep Reinforcement Learning (DRL) to mobile robot navigation has grown substantially following the success of DQN [11]. Tai et al. [10] demonstrated map-less navigation using continuous control, training policies in simulation and transferring to real robots. Zhu et al. [12] provide a comprehensive survey of DRL methods for navigation, highlighting challenges including sample efficiency and sim-to-real transfer.

For multi-robot systems, Everett et al. [22] proposed decentralized collision avoidance where each robot independently learns navigation policies, addressing the non-stationarity challenges of multi-agent RL. Long et al. [23] developed decentralized policies scalable to large robot teams. Specifically for leader-follower structures, prior work has applied DRL to train follower agents for trajectory tracking [24], shifting computational burden to the follower.

Our approach differs by assigning the learning-based navigation to the leader while employing simple proportional control for the follower. This hybrid strategy is particularly suitable for heterogeneous teams—such as “mother-ship” scenarios or convoys—where the leader possesses superior sensing and computing capabilities compared to followers. The decoupling simplifies the learning problem to single-agent RL while maintaining formation through robust classical control.

III. PROBLEM FORMULATION

Consider a two-robot system operating in a 2D workspace $\mathcal{W} \subset \mathbb{R}^2$ containing a set of known static obstacles $\mathcal{O} = \{O_1, \dots, O_k\}$. The system consists of a Leader (L) and a Follower (F), both modeled as differential-drive robots.

A. Formation Objective

Let the poses of leader and follower be $q_L = [x_L, y_L, \theta_L]^T$ and $q_F = [x_F, y_F, \theta_F]^T$, respectively. The formation objective requires the follower to maintain a desired Euclidean distance $d_{des} \in \mathbb{R}^+$ and relative bearing $\psi_{des} = \pi$ (directly behind) with respect to the leader (Figure 1).

The desired follower position, denoted P_{des} , is time-varying and depends on the leader’s current state:

$$P_{des}(t) = \begin{bmatrix} x_L(t) - d_{des} \cos(\theta_L(t)) \\ y_L(t) - d_{des} \sin(\theta_L(t)) \end{bmatrix} \quad (5)$$

The tracking error in the global frame is defined as:

$$\mathbf{e} = P_{des} - \begin{bmatrix} x_F \\ y_F \end{bmatrix} \quad (6)$$

Follower’s control objective is to drive $\|\mathbf{e}\| \rightarrow 0$ as $t \rightarrow \infty$.

B. Leader Navigation Problem

The leader operates as an RL agent navigating toward a goal position $G = [x_g, y_g]^T$ while avoiding obstacles. The state space $\mathcal{S}_L$ comprises:

- Relative polar coordinates to the goal: $(\rho_{goal}, \alpha_{goal})$
- Current velocities: v_L, ω_L

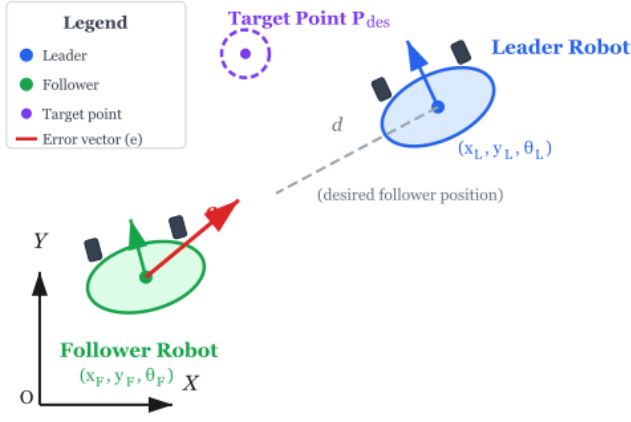


Fig. 1. Leader-follower formation geometry. The follower tracks a virtual target P_{des} located at distance d_{des} behind the leader along its heading direction. The error vector e represents the deviation to be minimized.

The action space $\mathcal{A}_L$ is a discrete set of admissible velocity pairs (v, ω) . The reward function balances goal-seeking and collision avoidance:

$$R_t = r_{goal} \cdot \mathbb{I}_{goal} + r_{coll} \cdot \mathbb{I}_{obs} + r_{dist}(\rho_{goal}) \quad (7)$$

where $\mathbb{I}_{goal}$ and $\mathbb{I}_{obs}$ are indicator functions for reaching the goal or colliding with obstacles, and r_{dist} provides dense feedback based on progress toward the goal.

C. System Constraints

Both robots are subject to actuator limits: $|v| \leq v_{max}$ and $|\omega| \leq \omega_{max}$. Inter-robot communication occurs via ROS 2 topics, introducing transport latency τ . Consequently, the follower receives delayed leader state $q_L(t - \tau)$, requiring the control law to be robust to reference signal delays.

IV. PROPOSED SOLUTION

This section presents the hybrid control architecture. The system decouples navigation complexity: the leader handles high-level decision-making and obstacle avoidance through a learned policy, while the follower executes deterministic tracking control. Figure 2 illustrates the overall system architecture.

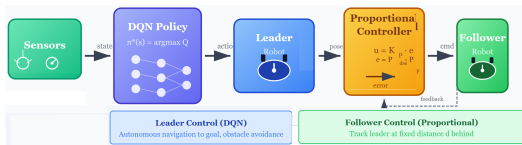


Fig. 2. System architecture for trained policy deployment. The leader processes sensor data through the DQN policy to generate motion commands, broadcasting its pose via ROS 2. The follower computes tracking error relative to the virtual target and applies proportional control.

A. System Architecture

The architecture comprises two subsystems operating in a decentralized manner for motion computation while relying on explicit communication for formation maintenance:

- 1) **Leader Subsystem:** It processes sensor data to estimate state s_t and queries the trained DQN to obtain action $a_t = (v_L, \omega_L)$.
- 2) **Follower Subsystem:** Operating as a slave node, the follower receives the leader's pose q_L via ROS 2 topics. To mitigate network latency and discrete sampling effects, we set up a delay of four steps to maintain the most recent leader state between updates. The follower keeps tracking the leader's orientation while maintaining a distance of d_{gap} m behind.

B. Leader Control Policy via DQN

The leader's navigation strategy is learned through Deep Q-Networks. Figure 3 illustrates the training process.

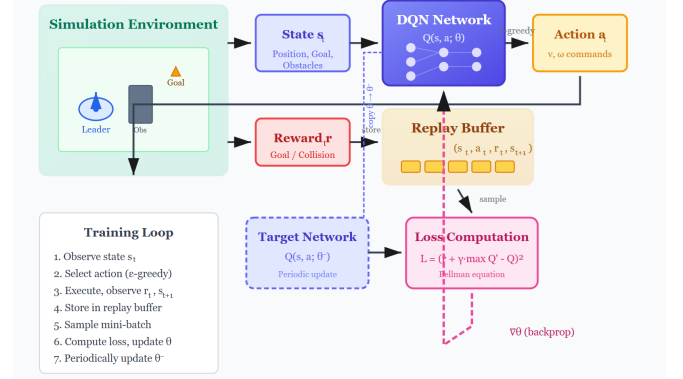


Fig. 3. DQN training. Agent observes state s_t , acts by a_t , and receives reward r_t . Transitions are stored in the replay buffer for network training.

- 1) **State Space:** To promote policy generalization, the state vector is defined in the robot's local frame:

$$s_t = [L_{goal_x}, L_{goal_y}, d_{goal}, pair, \sin(\theta_L), \cos(\theta_L), \sin(\theta_F), \cos(\theta_F), \psi_{head}]^T \quad (8)$$

where L_{goal_x} and L_{goal_y} are the error distance to the goal in x and in y , d_{goal} is the Euclidean distance do the goal, $pair$ is the Euclidean distance between the two robots, $\sin(\theta_L)$, $\cos(\theta_L)$, $\sin(\theta_F)$, $\cos(\theta_F)$ are the trigonometry decomposition of the robots angle with respect to x axis, and ψ_{head} is the relative heading.

- 2) **Action Space:** DQN requires discrete actions. We discretize the velocity space into K predefined values:

$$\mathcal{A} = \{(v_k, \omega_k) \mid k = 1, \dots, K\} \quad (9)$$

These include forward motion, combined translation-rotation, and pure rotation, enabling maneuverability in confined spaces.

- 3) **Network Architecture:** The action-value function $Q(s, a; \theta)$ is approximated by a fully connected Multi-Layer Perceptron (MLP) parameterized by θ . The network input dimension matches the state vector length, and the output dimension equals the number of discrete actions $|\mathcal{A}|$.

The MLP is composed of an input layer, two hidden layers, and an output layer. Let $h(\cdot)$ denote a nonlinear activation

function. The forward mapping is defined as

$$Q(s; \theta) = f_{\text{out}}(W_3 h(W_2 h(W_1 s + b_1) + b_2) + b_3), \quad (10)$$

where W_i and b_i denote the weight matrices and bias vectors of layer i , respectively, and $f_{\text{out}}(\cdot)$ is the output activation function.

Rectified Linear Units (ReLU) are employed as the hidden-layer activation function,

$$h(x) = \max(0, x), \quad (11)$$

due to their favorable optimization properties in deep reinforcement learning. The output layer uses an identity activation, producing a vector of Q-values,

$$Q(s; \theta) = [Q(s, a_1; \theta) \quad Q(s, a_2; \theta) \quad \cdots \quad Q(s, a_{|\mathcal{A}|}; \theta)]^T, \quad (12)$$

where each component corresponds to the estimated return of selecting action a_k in state s .

To improve training stability, a separate target network with parameters θ^- is maintained. While the online network parameters θ are updated at every optimization step, θ^- is updated only at discrete intervals, and is used to compute the bootstrap targets in the temporal-difference learning rule.

The network depth was intentionally kept shallow, as the state space is low-dimensional and the action space is discrete and limited, avoiding unnecessary representational complexity. This architectural choice ensures low inference latency, making the policy suitable for real-time deployment on computationally constrained robotic platforms.

4) *Reward Function*: It is designed to encourage goal-directed navigation, formation preservation, and safe behavior. At each time step t , the reward is defined as

$$r_t = r_{\text{prog}} + r_{\text{form}} - r_{\text{ang}} - r_{\text{lat}} - r_{\text{safe}} - r_{\text{time}}, \quad (13)$$

where:

- r_{prog} rewards progress toward the goal based on the reduction of d_{goal} ,
- r_{form} encourages maintenance of the desired inter-robot distance,
- r_{ang} penalizes heading misalignment,
- r_{lat} penalizes lateral deviation from corridor centerline,
- r_{safe} penalizes violations of minimum safety distances,
- r_{time} discourages unnecessarily long trajectories.

Sparse terminal rewards are assigned upon successful task completion, while large negative rewards are applied in the case of collisions, unsafe proximity between robots, or leaving the workspace boundaries.

C. Follower Control Law

The follower employs a deterministic proportional controller to track the virtual target generated by leader motion. Given leader pose $q_L = [x_L, y_L, \theta_L]^T$, the target position at distance d_{gap} behind is:

$$P_{\text{des}} = \begin{bmatrix} x_L - d_{\text{gap}} \cos(\theta_L) \\ y_L - d_{\text{gap}} \sin(\theta_L) \end{bmatrix} \quad (14)$$

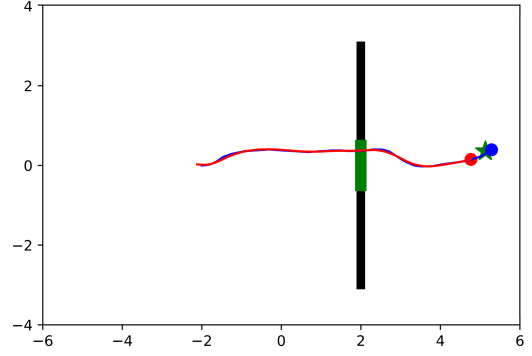


Fig. 4. Simulation scenario showing the leader (blue) and follower (red) robots navigating through a narrow passage (green region within the black wall) toward the goal (green star). The magenta trajectory shows the path taken by the formation.

The position error is transformed to the follower's local frame via rotation matrix $R(\theta_F)$:

$$\begin{bmatrix} e_x \\ e_y \end{bmatrix} = \begin{bmatrix} \cos(\theta_F) & \sin(\theta_F) \\ -\sin(\theta_F) & \cos(\theta_F) \end{bmatrix} \begin{bmatrix} x_{\text{des}} - x_F \\ y_{\text{des}} - y_F \end{bmatrix} \quad (15)$$

The control law computes velocity commands as:

$$v_F = k_v \cdot e_x, \quad \omega_F = k_\omega \cdot \arctan\left(\frac{e_y}{e_x}\right) \quad (16)$$

where $k_v, k_\omega > 0$ are gain constants. Computed velocities are saturated at $[v_{\text{max}}, \omega_{\text{max}}]$ to ensure physical feasibility. This control law drives the follower toward the virtual target while aligning its heading with the error vector direction.

Remark IV.1. The proportional controller (16) is computationally lightweight, requiring only basic arithmetic operations. This makes it suitable for resource-constrained follower platforms, concentrating computational demands on the leader where the DQN inference occurs.

V. SIMULATION RESULTS

This section presents simulation results validating the proposed hybrid control architecture. The experiments evaluate the leader's learned navigation policy and the follower's tracking performance across scenarios of varying difficulty.

A. Experimental Setup

The simulation environment consists of a 2D workspace with a wall obstacle containing a narrow passage of 1 m width, as illustrated in Fig. 4. The leader robot must navigate from the left side of the wall to reach a goal position on the right side, while the follower maintains formation. The episode terminates successfully when the leader reaches within 0.3 m of the target.

The follower robot is initialized within a 1 m radius of the leader's starting position and must maintain approximately 0.3 m distance behind the leader throughout the navigation task. The controller is pure proportional to the error of distance and angle, with $k_{\text{dist}} = 1$ and $k_{\text{angle}} = 1.4$.

The action space was made very small, with $K = 3$, where:

$$\mathcal{A}_{\square} = \{0.00, 0.25, 0.50\}, \quad (17)$$

$$\mathcal{A}_\omega = \{-0.7, 0.0, 0.7\} \quad (18)$$

The DQN policy was trained using the parameters summarized in Table I.

TABLE I
TRAINING AND SIMULATION PARAMETERS

Parameter	Value
Discount factor γ	0.99
Learning rate α	1×10^{-3}
Replay buffer capacity	100,000
Mini-batch size	64
Target network update interval	1500 steps
Exploration decay λ	0.9993
Minimum exploration $\epsilon_{\min}$	0.05
Maximum episode length	300 steps
Formation distance d_{gap}	0.30 m
Follower gains (k_d, k_ω)	(1.0, 1.4)
Goal tolerance	0.10 m

B. Evaluation Scenarios

To assess the generalization capability of the learned policy, we evaluated the system under three initial configurations of increasing difficulty:

- **Easy:** The leader starts at $(-2, 0)$, directly aligned with the passage. This configuration requires minimal maneuvering—the leader essentially moves forward through the opening.
- **Medium:** The leader starts at $(-2, -0.8)$, slightly offset from the passage centerline. The policy must perform a gradual correction to align with the opening.
- **Hard:** The leader starts at $(-2, -1.2)$, positioned nearly parallel to the wall. This scenario is the most challenging because a direct path to the passage would cause the follower to collide with the wall edge.

C. Trajectory Analysis

Figure 5 presents the trajectories for all three scenarios. In the Easy case, the leader (blue) follows a nearly straight path to the goal, with the follower (orange) tracking closely behind. The Medium scenario shows a gentle corrective maneuver as the gray leader adjusts its heading toward the passage. The Hard scenario reveals an interesting behavior: rather than taking the shortest path directly toward the passage, the learned policy commands the leader (red) to first move upward, aligning itself perpendicularly to the wall before proceeding through the opening. This maneuver creates sufficient clearance for the follower (green) to maintain the prescribed formation distance without colliding with the passage edges. This behavior was not explicitly programmed but emerged from the training process, demonstrating the policy’s ability to account for formation constraints.

The Hard scenario reveals an interesting behavior: rather than taking the shortest path directly toward the passage, the learned policy commands the leader (red) to first move upward, aligning itself perpendicularly to the wall before proceeding through the opening. This maneuver creates sufficient clearance for the follower (green) to maintain the prescribed formation distance without colliding with the passage edges. This behavior was not explicitly programmed but emerged from the training process, demonstrating the policy’s ability to account for formation constraints.

D. Goal Convergence

Figure 6 shows the evolution of the Euclidean distance from the leader to the goal over time for all three scenarios. All trajectories exhibit monotonic convergence toward the target, confirming successful goal-reaching behavior.

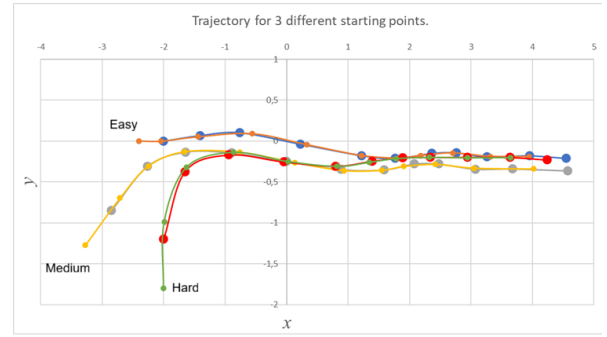


Fig. 5. Leader-follower trajectories for three initial conditions: Easy (blue/orange), Medium (gray/yellow), and Hard (red/green). The learned policy exhibits emergent behavior in the Hard case, where the leader first aligns perpendicularly to the passage before approaching, ensuring the follower maintains formation without collision.

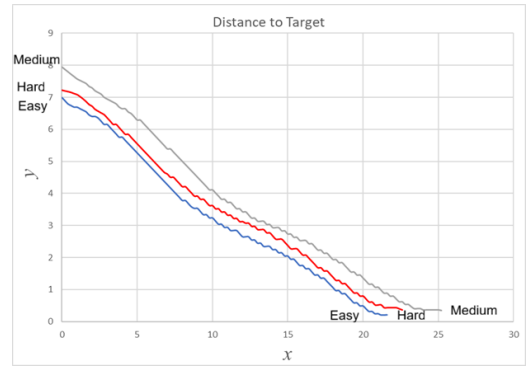


Fig. 6. Distance to target over time for the three evaluation scenarios. The Easy case (blue) converges fastest due to the shorter initial distance and direct path. The Medium (gray) and Hard (red) cases require additional time steps to complete the navigation task due to the corrective maneuvers needed to traverse the passage safely.

The Easy configuration achieves the fastest convergence, reaching the goal vicinity in approximately 20 time steps, which is expected given the shorter initial distance (≈ 7 m) and direct trajectory. The Medium case requires roughly 25 steps, starting from a greater initial distance (≈ 8 m) and executing a mild heading correction. The Hard scenario, despite starting at a similar distance to the Medium case, takes the longest to complete due to the deliberate detour maneuver that prioritizes formation safety over path length.

E. Discussion

The simulation results demonstrate several key aspects of the proposed architecture:

Policy Generalization: The DQN-trained leader successfully navigates to the goal from diverse initial conditions without retraining, indicating good generalization of its policy.

Emergent Coordination: Although the leader’s policy was trained independently (single-agent RL), it learned behaviors that implicitly facilitate follower tracking, as evidenced by the perpendicular alignment maneuver in the Hard scenario.

Formation Maintenance: The proportional controller enables the follower to track the virtual target point effectively across all scenarios, maintaining the prescribed formation distance

even during complex leader maneuvers.

1) *Narrow Passage Navigation*:: The system successfully coordinates both robots through the 0.8 m passage while maintaining approximately 0.5 m formation distance, demonstrating the viability of the hybrid approach for constrained environments.

These results validate the effectiveness of combining learning-based navigation for the leader with classical control for the follower, achieving adaptive obstacle avoidance while maintaining formation integrity.

VI. CONCLUSION

This paper presented a hybrid control architecture for leader-follower formation control of non-holonomic mobile robots in obstacle-rich environments. By combining Deep Q-Network-based navigation for the leader with classical proportional control for the follower, the approach effectively decouples the complexity of obstacle avoidance from formation maintenance.

The proposed architecture was validated in a challenging scenario involving navigation through a narrow 0.8 m passage while maintaining 0.5 m formation distance. Simulation results across three initial configurations of varying difficulty demonstrated consistent goal convergence for all cases. Notably, the learned policy exhibited emergent coordinated behavior in the most challenging scenario: rather than pursuing the shortest path, the leader autonomously executed a perpendicular alignment maneuver to create sufficient clearance for the follower to traverse the passage without collision. This behavior emerged naturally from the single-agent training process, suggesting that the DQN implicitly learned to account for formation constraints.

The hybrid strategy proved particularly effective for scenarios where computational resources are unevenly distributed between robots, concentrating the learning-based decision-making on the leader while enabling lightweight followers to maintain formation through simple proportional control. Future work will extend this framework to multiple followers with dynamic formation reconfiguration, investigate robustness to communication delays and packet loss, and validate the approach through hardware experiments on physical robot platforms.

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